

Forensic Microscopy

Lab 7: Contrast Techniques

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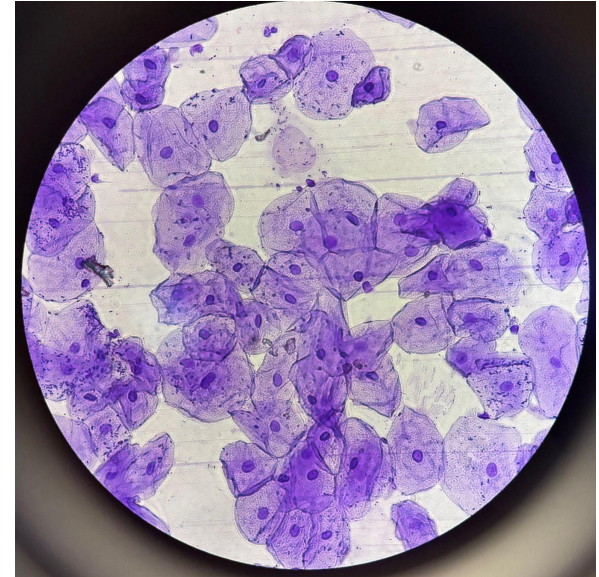
Where can you find the slides?



- <https://aritra-evolves.github.io/FIVS215.html>
- Canvas

Agenda

- ❖ What are contrast techniques?
- ❖ Types, examples, applications
- ❖ Practical



Five Types

Phase Contrast Microscopy (PCM)

- ❖ Diffracted and non-diffracted light is shifted to improve constructive & destructive interference conditions
- ❖ Uses different phase rings
 - Need to be aligned in order to work properly

Brightfield and Darkfield

- ❖ Focuses light centrally or externally
 - Photos on the lecture slides explain it well

Dispersion Staining (DS)

- ❖ Uses central & annual stops to create colors
- ❖ Primarily used to identify spermatozoa in sexual assault cases - Christmas Tree Stain

Modulation Contrast

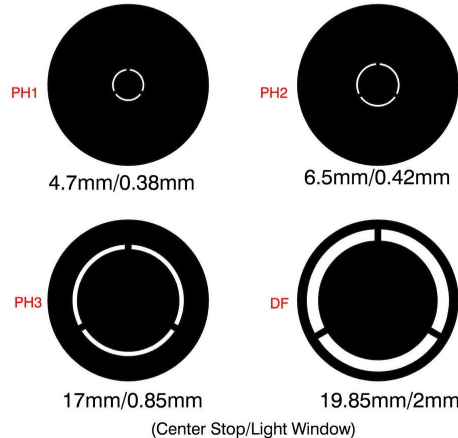
- ❖ Uses oblique illumination to create a shadowing effect
- ❖ a.k.a. Hoffman Modulation

How does Phase contrast work?

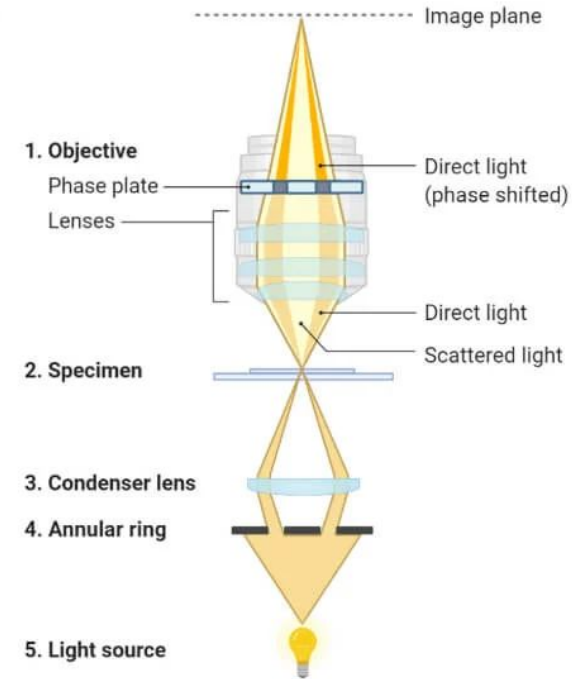
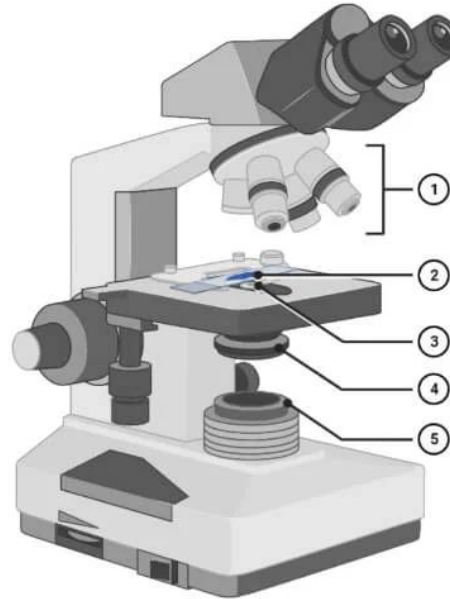


How does Phase contrast work?

LEITZ Microscope Phase Contrast Annulus Set 27mm Diameter



Phase Contrast Microscopy

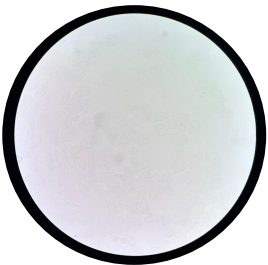


Bright field, dark field & phase contrast

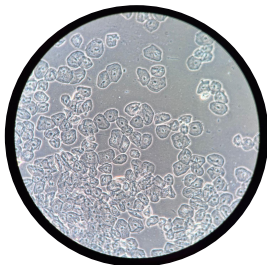
Bright field: Mostly for stained samples.

Dark field: Mostly for unstained Samples.

Phase Contrast: To see architecture of unstained samples that has little contrast in bright field.



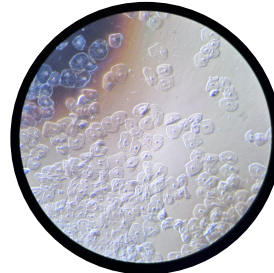
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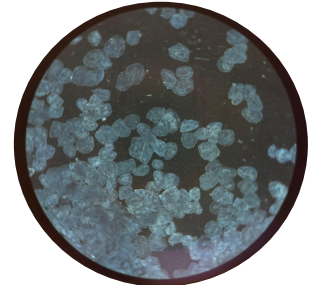
Ph1



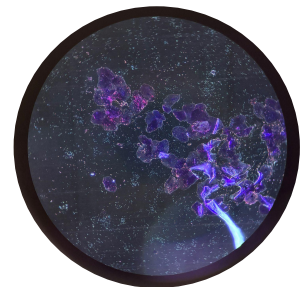
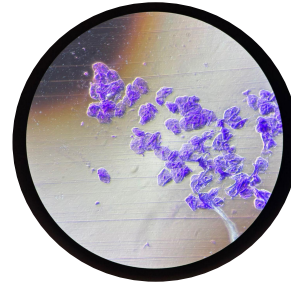
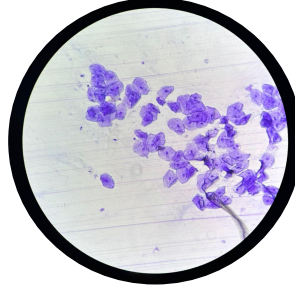
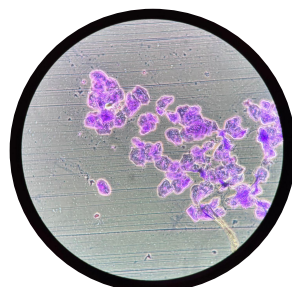
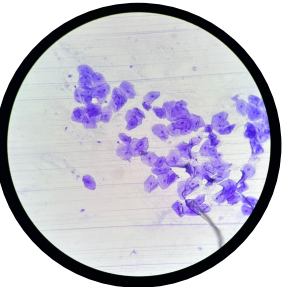
Ph2



Ph3



DF



Dispersion Staining

- Dispersion staining is a particle identification technique based on the difference between refractive index dispersion of a particle and of the liquid medium in which the particle is immersed.
- Two different procedures are based on the use of stops in the objective back focal plane; both give colored particle boundaries.

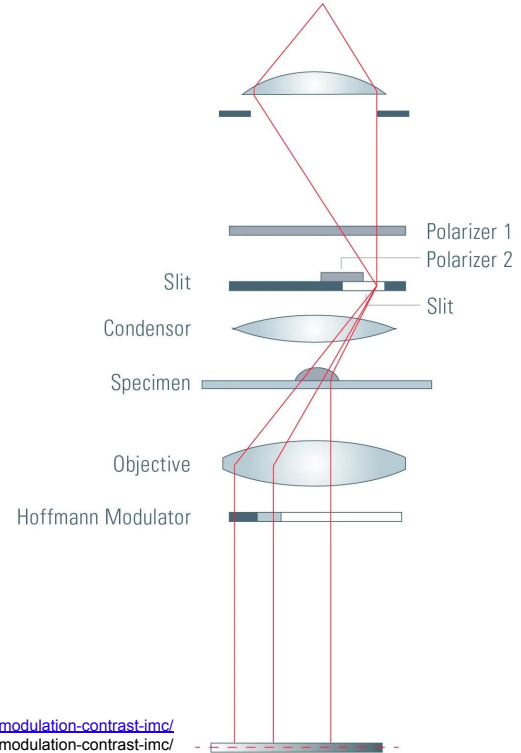
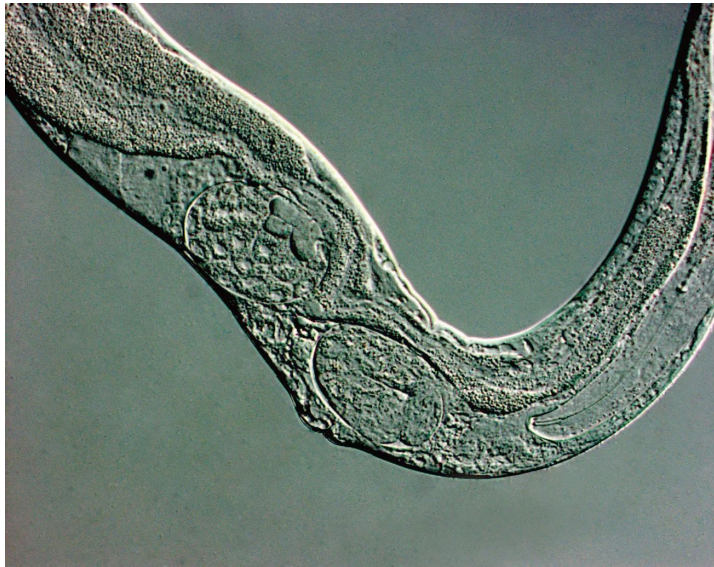
Take a look at the Turret



<https://mccroneinstitute.org/the-microscope-journal/the-microscope-70-3-2023/>

(Hoffman) Modulation Contrast

- ❖ Uses oblique illumination to create a shadowing effect
- ❖ Different sizes of slits filter light at angles
- ❖ Drastically increases resolution
- ❖ Creates a 3d like image



<https://www.leica-microsystems.com/science-lab/microscopy-basics/integrated-modulation-contrast-imc/>
<https://www.leica-microsystems.com/science-lab/microscopy-basics/integrated-modulation-contrast-imc/>

Christmas tree staining

Christmas tree staining is used in forensics to investigate sexual assault.

This method stains the sperm head red and the tail green, which makes a clear contrast even when there are debris.

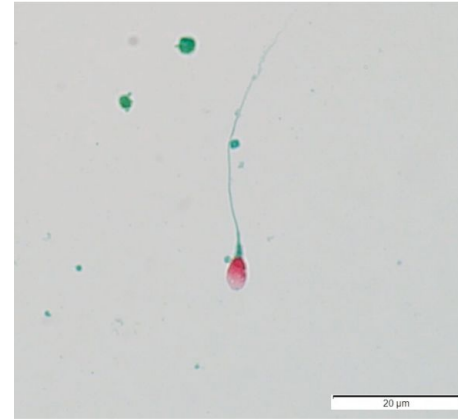


FIGURE 1—Cytology of human sperm



<https://serological.com/forensic-dna-testing/facility-items/christmas-tree-stain-75/>

<https://www.semanticscholar.org/paper/Sperm-cytology%3A-Kernochrot%2E%80%9393picroindigocarmine-and-Sperm/d1b99d8b425a43b9d48b92b3d147e56c196c6051>

For Today

1. Focus on your Specimen on 10X
2. Open the Aperture all the Way
3. Observe Brightfield
4. Swing in the Phase 1 Ring
5. Observe the Back Focal Plane by Removing an Eyepiece
6. Center the Ring of Light
7. Add a Filter
8. Observe the Colors in Orthoscopic Viewing (with the Eyepieces)
9. View with Darkfield (DF)
10. Repeat on 20X (Ph1) and 40X (Ph2)



In Your Notebook

- ❖ Tell me what you saw
- ❖ Tell me how rotating the stage changed what you saw

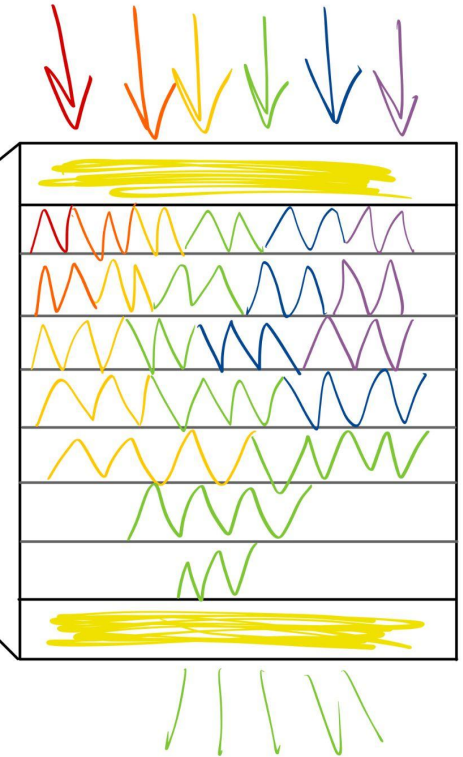
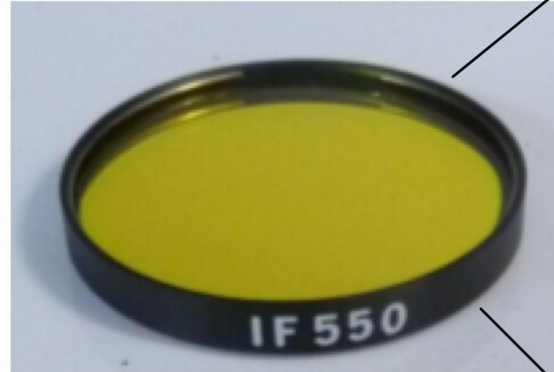
Interference Figures

Utilizes Mirrors

Limit Wavelengths to VERY Narrow Band

Increase Resolution

More Precise than Monochromatic Filters



Turret



Image: <https://spectraservices.com/product/6-1p151a.html?srsltid=AUI7owdVhTcZBilHwKSBjG9DRYPEfJEin6d72l25qhdgOK961BjVSYdHj>
<https://www.labx.com/item/mccrone-microscope-dso-dispersion-staining-objective-10x/DIS-88306-fd00633fd6fc7ce427d7a32e86e065c6>

See you next week!